ESC194: Calculus I
Exam Review

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3 months of Calculus I in 4 pages.

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1 Limits and Continuity

1.1 The Precise Definition of a Limit

The precise definition of a limit is as follows.

Definition. If $\forall \epsilon > 0$, $\exists \delta > 0$ such that $|x - a| < \delta \implies |f(x) - L| < \epsilon$, then

$$\lim_{x \to a} f(x) = L.$$

For one-sided limits, x is only approaching a from one side. Thus, we define the left-handed limit as follows.

Definition. If $\forall \epsilon > 0$, $\exists \delta > 0$ such that $a - \delta < x < a \implies |f(x) - L| < \epsilon$, then

$$\lim_{x \to a^-} f(x) = L.$$

As a number becomes arbitrarily large, it eventually surpasses every real number. Thus, we define limits to or approaching infinity as follows.

Definition. If $\forall M > 0$, $\exists \delta > 0$ such that $0 < |x - a| < \delta \implies f(x) > M$, then

$$\lim_{x \to a} f(x) \to \infty.$$

Conversely, if $\forall \epsilon > 0$, $\exists M > 0$ such that $x > M \implies |f(x) - L| < \epsilon$, then

$$\lim_{x \to \infty} f(x) = L.$$

Question. *How can we rigorously define right-handed and negative infinite limits? What about limits involving both concepts?*

Here are the general steps to a rigorous proof.

1. Given $\epsilon > 0$, find $\delta > 0$. Start with $f(x) - L < \epsilon$ and manipulate it into the form $|x - a| < \delta_0$. This step may require bounding $\delta \leq n$.
2. *Proof.* Given ϵ, let $\delta = min(n, \delta_0)$. Then, show that $|x - a| < \delta \implies |f(x) - L| < \epsilon$, as desired. □

1.2 Limit Theorems

Here are some important limit theorems.

Theorem 1.1. *If* $\lim_{x \to a} f(x)$ *and* $\lim_{x \to a} g(x)$ *exist, then*

- *(Constant LT)* $\lim_{x \to a} c = c$
- *(Sum LT)* $\lim_{x \to a}[f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$
- *(Product LT)* $\lim_{x \to a}[f(x) \cdot g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$
- *(Constant multiple LT)* $\lim_{x \to a} cf(x) = c \lim_{x \to a} f(x)$

- *(Power LT)* $\lim_{x\to a}[f(x)]^n = [\lim_{x\to a} f(x)]^n$
- *(Root LT)* $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$

Theorem 1.2 (Direct substitution)**.** *If f is continuous at a, then*

$$\lim_{x\to a} f(x) = a.$$

The following theorem is often used to evaluate trigonometric limits.

Theorem 1.3 (Squeeze theorem)**.** *If $g(x) < f(x) < h(x)$ and $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$, then*

$$\lim_{x\to a} f(x) = L.$$

When solving difficult trigonometric limits, it may help to find $\frac{\sin x}{x}$ terms.

1.3 Continuity Theorems

Here is the definition for continuity.

Definition. f is continuous at a point a if $\lim_{x\to a} f(x) = f(a)$.

Question. *How can we define continuity from the side? What about continuity on an open or closed interval?*

Here are some continuity theorems.

Theorem 1.4. *If f and g are continuous at a, then*

$$f + g,\ fg,\ cf$$

are also continuous at a.

Theorem 1.5. *The following types of functions are continuous on their domains*

- *polynomial*
- *rational*
- *root*
- *exponential*
- *logarithmic*
- *trigonometric*

In fact, all elementary functions are continuous on their domains.

2 Derivatives

2.1 Derivative Theorems

We begin with the definition of the derivative.

Definition. The derivative of f is defined as

$$f'(x) \coloneqq \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Now we define differentiability.

Definition. f is differentiable at a point a if $f'(a)$ exists.

Question. *How can we define differentiability from the side? What about differentiability on an open or closed interval?*

From the definition of the derivative, we see that differentiability at a requires $\lim_{h \to 0}[f(a+h) - f(a)] = 0$ (why?). This implies that f is continuous at a. Thus, we arrive at the following theorem.

Theorem 2.1. *If f is differentiable at a, then f is continuous at a.*

Be careful, as this theorem does not hold for the converse. That is, continuity at a does not imply differentiability at a.